



Change the Event

Change a hat and describe what changes

Name: _____

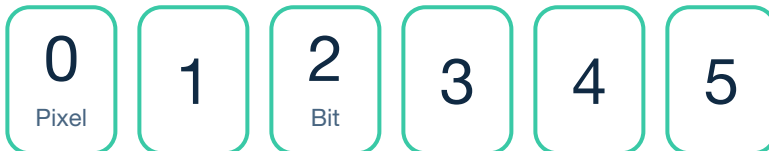
Date: _____

★ I can change a script's event and describe how it changes when the script runs.

Change the hat, change the timing

Bit starts on the green flag and broadcasts 'go'. Right now Pixel's hat is 'when I receive go', so Pixel WAITS for the message and runs second. We will CHANGE Pixel's hat to 'when green flag clicked'. Watch how that changes WHEN Pixel runs.

The number path



Bit's script

when green flag clicked

forward 2

broadcast 'go'

Pixel's script NOW

when I receive 'go'

forward 1

1 Before the change

Run it now. Which sprite runs FIRST and which runs SECOND? Why does Pixel run second?

The change

Now change Pixel's top hat from 'when I receive go' to 'when green flag clicked'. Bit's script stays the same. Here is Pixel's NEW script:

Pixel's script AFTER the change

🚩 when green flag clicked

forward 1

2 After the change

Now when does Pixel run — does it still wait for the message, or does it start at the same time as Bit?
Circle: WAITS / STARTS WITH BIT.

3 Describe the change

Write one sentence: how did changing Pixel's event change WHEN Pixel runs? (Both still land the same: Bit 4, Pixel 1.)

Big idea

The event hat decides WHEN a script runs. Changing Pixel's hat from a message to the green flag makes Pixel start at the same time as Bit (parallel) instead of waiting for the message (second). The landing numbers stay the same — the timing changes.



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Quick facilitation notes & answer key

What this worksheet teaches

Students change which EVENT starts a set of moves and describe the effect. Swapping Pixel's hat from "when I receive go" to "when green flag clicked" changes Pixel from running SECOND (waiting for the message) to running at the SAME TIME as Bit. No coding experience needed.

Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: two counters on a number line 0 to 5 (green Bit, purple Pixel), and a card that says "go" you can hand over to act out sending a message.

How to lead it (about 20 minutes)

1. Show them (you do). Run it as is: Bit first, Pixel second (it waits for the message).
2. Do one together. Exercise 1 — record who runs first and second and why.
3. Let them try. Students run with Pixel's new green-flag hat (Ex 2), then write how the event change changed WHEN Pixel runs (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 (before) — Bit runs first (green flag); Pixel runs second because it waits for the "go" message.

Exercise 2 (after) — with the green-flag hat, Pixel no longer waits; it starts at the same time as Bit.

Exercise 3 — sample: "Changing Pixel's hat to the green flag made Pixel start at the same time as Bit instead of waiting for the message." The landings stay the same (Bit 4, Pixel 1); only the timing changed.

If students get stuck — and ways to adjust

Watch for: thinking the landing numbers change. Only WHEN Pixel starts changes, not where it lands.

Make it easier: compare "who starts first" before and after the change.

Make it harder: ask what hat would make Pixel run THIRD, after another message.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions that are started by an event, like clicking the green flag or getting a message.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, change what starts them or one move, and explain what the change did.

You'll know it worked when

The child changes Pixel's starting event and explains how it changed the timing.